

Problem 1 (§14.1, #8). Show that $\mathbb{Z}$ and $S = \{x \in \mathbb{R} : \sin(x) = 1\}$ have equal cardinality by describing a bijection from one to the other.

Problem 2 (§14.2, #2). Prove that the set $A = \{(m, n) \in \mathbb{N} \times \mathbb{N} : m \leq n\}$ is countably infinite.

Problem 3 (§14.2, #4). Prove that the set of all irrational numbers is uncountable.

Problem 4 (§14.3, #8). Prove or disprove: The set $\{(a_1, a_2, a_3, \dots) : a_i \in \mathbb{Z}\}$ of infinite sequences of integers is countably infinite.

Problem 5 (§14.4, #2). Show that $|\mathbb{R}^2| = |\mathbb{R}|$.